

Total no. of Pages: 3

Roll no.

Degree: B.Tech

Semester: VIth Course work

END-SEMESTER EXAMINATION, APRIL-MAY, 2024

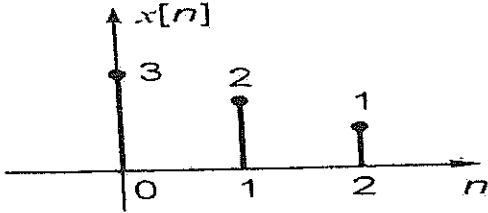
Course Title: Signal And Systems

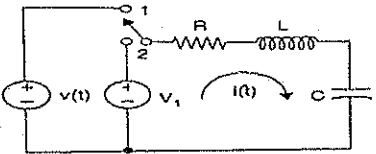
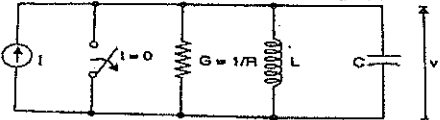
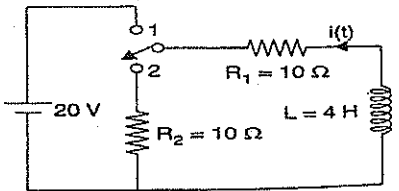
Course Code: EEEEC03

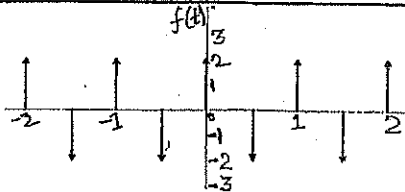
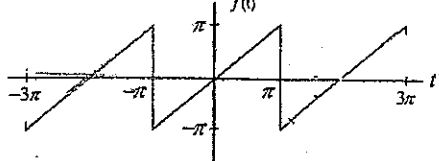
Duration: 03 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Discuss the classification of signals	5	1
1b	Calculate the period of composite signal $x[n]$. $x[n] = \sin\left(\frac{\pi}{3}n\right) + \cos\left(\frac{\pi}{4}n\right)$	5	1
1c	Draw the even and odd component of the waveform of signal $x[n]$ shown in the Figure 1. 	5	1
	Figure1.		
Q 2	Attempt any 2 parts of the following		
2a	What is the Laplace transform of $x(t) = -e^{2t}u(t) * t u(t)$	5	2
2b	Solve the inverse Laplace transform of system function $H(s) = \frac{1}{s^2 + 5s + 6}$ For three cases of ROCs when i. $\text{Re}\{s\} > -2$ ii. $\text{Re}\{s\} < -3$ iii. $-3 < \text{Re}\{s\} < -2$	5	2
2c	Evaluate the Laplace transform of the signal and corresponding ROCs $x(t) = e^t u(t) + e^{-2t} u(-t)$	5	2
Q 3	Attempt any 2 parts of the following		
3a	Consider the circuit shown in the Figure 2. The switch is thrown from position 1 to 2 at time $t=0$. Just before the switch	5	3

	is thrown the initial conditions are $i_L(0^+) = 2A$, $v_C(0^+) = 2V$. Find the current $i(t)$ after the switch is thrown. Assume $L = 1H$, $C = 0.5F$, $V_1 = 5V$.  Figure 2.		
3b	Consider a parallel RLC circuit as shown in Figure 3, the switch is opened at $t=0$. Determine the terminal voltage (v).  Figure 3.	5	3
3c	The circuit in Figure 4. Is initially under steady state condition. The switch is moved from position 1 to position 2 at $t=0$. Find the current after switching.  Figure 4.	5	3
Q 4 Attempt any 2 parts of the following			
4a	Find the Z-transform of the signal $x[n] = a^n \cos(\omega n) u[n]$	5	4
4b	Consider $X[z] = \frac{z^2}{(z-1)(z-0.5)}$ Evaluate $x[n]$ if ROCs are - ROC: $z > 1$ - ROC: $z < 0.5$ - ROC: $0.5 < z < 1$	5	4
4c	Define the Sampling Theorem and Nyquist criteria.	5	4
Q 5 Attempt any 2 parts of the following			
5a	Solve the Fourier Series of the signal $f(t)$ in Figure 5.	5	5

	 Figure 5		
5b	Find the Fourier Series of the signal $f(t)$ in Figure 6.	5	5
	 Figure 6		
5c	Describe the LTI system in detail.	5	5